

ANALYSTLAB AFRICA

Experience Lab Internship Programme

WEEK 1 SUBMISSION

FinTrust Digital Banking Customer & Transaction Intelligence

Data Analytics Track

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Week 1 - Understand & Plan

1. Summary

FinTrust Digital Bank is seeking to use customer and transaction data to improve its understanding of customer and transaction behaviour, support risk-related decision-making, improve operational processes and strengthen digital customer support. As a Data Analytics intern, my role is to transform the supplied customer and transaction data into useful business intelligence.

The proposed Week 1 analytical focus is to understand how banking activity varies across customers, channels, devices, transaction types and locations. Particular attention will be given to the distinction between transaction volume and unique-customer usage so that high activity is not automatically interpreted as widespread customer adoption without considering other smaller intricate details.

Proposed core analytical question

How does digital banking usage vary across customers, channels, devices and locations, and what transaction and operational patterns can FinTrust identify from this behaviour?

2. Business Understanding

2.1 What problem is FinTrust trying to solve?

FinTrust has customer and transaction data, but management needs structured analysis to turn that data into actionable business intelligence. The key information gap addressed by this project is visibility into how customers interact with the banking ecosystem: which channels and devices they use, how activity differs across customer segments and locations, what transaction types and values dominate, and where transaction outcomes or risk-review patterns warrant further investigation.

2.2 Why is the problem important?

Understanding customer and transaction behaviour can support evidence-based decisions about digital-channel planning, customer engagement, service improvement and operational monitoring. It can also help FinTrust distinguish between broad adoption and activity only concentrated among a smaller number of frequent users.

2.3 What business questions should FinTrust's management be able to answer?

- How does digital banking engagement vary across customer characteristics such as age, income band, customer segment and account type?
- How is transaction activity distributed across the geographic locations represented in the dataset?
- Which channels and transaction types have the highest proportions of successful, failed, pending or reversed transactions?
- Are customers who have higher digital engagement scores also more active in digital banking transactions?
- How many customers are actively using FinTrust's digital banking services?
- How does digital banking engagement vary across customer characteristics such as age, income band, customer segment and account type?
- Are there particular customer, transaction, channel, device or geographic characteristics associated with transactions flagged for risk review?

The business questions focus on understanding who is using FinTrust's services, how they are using them, where activity is concentrated, what transactions are taking place, and where operational or risk-related patterns may exist. The resulting analysis can support decisions relating to digital-channel investment, customer engagement, platform optimisation, operational improvement, geographic planning and risk monitoring.

2.4 How can Data Analytics contribute?

- Profile the supplied customer and transaction datasets and establish their structure and data.
- Combine customer data with transaction data using Customer_ID.
- Measure channel, device, transaction, geographic and customer-segment patterns.
- Develop clearly defined Key Performance Indicators that management can monitor.
- Translate findings into an interactive dashboard which gives concise business insights.
- Identify patterns for further investigation.

2.5 Potential stakeholders

Stakeholder	Potential analytical value
Management / Leadership	Understand overall customer behaviour, digital usage and operational performance.
Digital Banking / Product Team	Understand channel and device usage to support digital-service planning.
Customer Experience / Support	Identify patterns in transaction outcomes and channel usage that may affect customers.
Operations Team	Monitor transaction success, failure and reversal patterns.
Risk / Compliance Team	Explore patterns associated with the synthetic Risk_Review_Flag.
Data / Analytics Team	Use the analysis as a reproducible foundation for further modelling or reporting.

3. Data Understanding & Initial Profiling

3.1 Dataset summary

Dataset	Records	Columns	Dictionary fields	Missingness	Initial duplicate check
Customer dataset	1,500	12	12	No missing values	No duplicate rows or Customer_IDs

Transaction dataset	12,000	11	11	192 missing cells	No duplicate rows or Transaction_IDs
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3.2 Customer data profile

Type	Fields	Analytical relevance
Text / identifiers	Customer_ID, Customer_Name	Customer_ID is the primary key; Customer_Name is display/identification data.
Categorical	Gender, City, Customer_Segment, Account_Type, Monthly_Income_Band, Preferred_Channel, Account_Status	Suitable for segmentation and comparison.
Numerical	Age, Tenure_Months, Digital_Engagement_Score	Suitable for demographic, lifecycle and engagement analysis.

Observed customer ranges include Age 18–65, Tenure_Months 1–96, and Digital_Engagement_Score 16.7–100. The dataset contains eight cities, four customer segments, three account types, five income bands, three preferred channels and three account-status categories.

3.3 Transaction data profile

Type	Fields	Analytical relevance
Identifiers / relationship	Transaction_ID, Customer_ID	Transaction_ID identifies each transaction; Customer_ID links transactions to customers.
Date/time	Transaction_DateTime	Supports trend, time-of-day and period analysis.
Categorical	Transaction_Type, Channel, Device_Type, Location, International_Transaction, Transaction_Status, Risk_Review_Flag	Supports transaction mix, channel/device, geographic, operational and risk-pattern analysis.
Numerical	Amount_NGN	Supports transaction value, average/median value and anomaly-oriented analysis.

The transaction period runs from 1 January 2026 to 31 March 2026. Transaction amounts range from NGN 100.17 to NGN 693,454.45. The total recorded transaction value is NGN 560,477,354.85; the mean is NGN 46,706.45 and the median is NGN 10,305.94.

3.4 Missing values and obvious data-quality observations

Check	Result	Implication
Customer dataset	0 missing cells	No missing values observed in initial profiling.
Transaction Device_Type	96 missing (0.80%)	Missing device information may affect device-level comparisons.
Transaction Location	96 missing (0.80%)	Missing location information may affect geographic comparisons.
Duplicate Customer_IDs	0	No duplicate customer keys observed.
Duplicate Transaction_IDs	0	No duplicate transaction keys observed.
Orphan transaction Customer_IDs	0	All transaction Customer_IDs match the customer dataset.

4. Resource Review

Resource	What it contains	Expected use	Limitations
FinTrust_Customer_Data.xlsx	1,500 customer records; 12 fields covering demographics, location, customer segment, account information, digital engagement, income band, preferred channel and account status.	Customer profiling, segmentation and joining customer attributes to transactions.	Customer Income Band in a single column will make it difficult to perform some arithmetic analysis as there are non-numerical characters. Digital engagement should all ought to be at 1 decimal point.
FinTrust_Transaction_Data.xlsx	12,000 transaction records; 11 fields covering date/time, transaction type, amount, channel, device, location, international status, transaction status and risk-review flag.	Transaction timing particulars, channel identification on device, value, operational and risk-pattern analysis.	Only January-March 2026 is covered. Device_Type and Location each have missing values. Device data is transaction-level.
FinTrust_Data_Dictionary.xlsx	Field definitions, data types, examples, business uses, modelling uses and project notes.	Interpret field meaning and avoid incorrect assumptions about variables.	The Dictionary cannot supply information absent from the underlying datasets.

4.1 Data Dictionary: Further limitation on the Example column

The Example column in the Data Dictionary is not the data to be cleaned. It is an illustrative example of what a field value may look like. For example, Customer_ID has an example such as FT-C00001, while Age has an example such as 34. Data cleaning should be performed on the actual Customer and Transaction datasets after profiling, not on the Dictionary's Example column.

5. Data Analytics Track Objectives

1. Build a reliable understanding of the structure and relationships within the supplied customer and transaction datasets.
2. Develop an analytical framework for measuring customer, channel, device, geographic, transaction and operational behavior.
3. Produce 4-5 clearly defined KPIs and an evidence-based set of analytical insights.
4. Develop a reproducible Python/Pandas analysis.
5. Document limitations and assumptions so that conclusions remain proportional to what the supplied data can support.

6. Success Criteria

Success area	Evidence of success
Data integrity	Customer and transaction data are correctly profiled, joined and documented, with data-quality issues explicitly recorded.
Question coverage	The final analysis provides evidence-based answers to all questions.
KPI accuracy	KPI definitions are clear, reproducible and traceable to the existing data sets.
Business usefulness	Management can identify meaningful patterns in customer, channel, device, geographic and transaction behavior.
Reproducibility	The Python/Jupyter workflow can be rerun and produce the same analytical outputs from the same input data.
Analytical discipline	The analysis distinguishes descriptive patterns from causal claims and clearly states

	dataset limitations.
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7. Initial Week 2–4 Plan

Week	Planned activities	Expected output
Week 2 - Data Preparation & Exploratory Analysis	• Load and profile both datasets in Jupyter. • Investigate missing Device_Type and Location values. • Validate identifiers, duplicates and data types. • Define analysis-ready variables and join customer/transaction data. • Conduct exploratory analysis of customer, channel, device, location and transaction behavior.	Clean analysis dataset
Week 3 - KPI & Deeper Analysis	• Filter out approved KPIs. • Compare transaction volume with unique-customer usage. • Segment results by customer segment, age, income band, account type and location. • Analyze transaction status and synthetic risk-review patterns. • Convert analytical findings into concise business insights.	Approved KPI results; analytical findings; insight summary.
Week 4 - Dashboard & Final Business Intelligence Output	• Build the Power BI dashboard using validated metrics. • Add filters and clear visual hierarchy. • Validate dashboard figures against Python/SQL outputs. • Document assumptions, limitations and recommendations for further analysis. • Package the final work professionally for portfolio use.	Final dashboard; documented analytical workflow; final insight report; portfolio-ready project.

8. Analytical Questions

The assignment requires 5–8 meaningful analytical questions. The following eight questions cover customer behaviour, transaction activity, transaction value, channels, transaction status and risk-related patterns.

Question	Analytical question
Q1 Digital adoption	What proportion of FinTrust customers actively use digital banking channels, and how does digital engagement vary across customer segments?
Q2 Channel usage	Which banking channels are most frequently used by customers, and how does channel usage vary by customer segment?
Q3 Device usage	Which devices are most commonly used for transactions, and how does device usage differ in terms of transaction volume and number of unique customers?
Q4 Geographic activity	How does transaction activity vary across the geographic locations represented in the dataset?
Q5 Customer behavior	How do age, income band, customer segment and account type relate to digital banking engagement and transaction behaviour?
Q6 Transaction value	Which transaction types and channels generate the highest transaction values?
Q7 Transaction performance	Which transaction channels and transaction types have the highest rates of failed or reversed transactions?
Q8 Risk patterns	What customer, transaction, channel, device and geographic characteristics are associated with transactions flagged for risk review?

8.1 Important analytical definitions

- Transaction volume and customer adoption will be measured separately. A large number of transactions does not necessarily mean a large number of unique users.
- For initial planning, digital channels are proposed as Mobile App, Web and USSD. This definition will be validated against the project data and business context during Week 2.
- The available location field contains eight cities. The dataset does not contain a rural/town/village classification, so geographic analysis will be limited to the locations actually represented.
- Risk_Review_Flag will be treated as a synthetic educational label. It will not be described as proof of fraud or wrongdoing.

9. KPI Definition Table

KPI	Definition	Why it matters	Required data
Total Transactions	Count of transaction records.	Measures overall transaction activity.	Transaction_ID
Total Transaction Value	Sum of Amount_NGN.	Measures monetary activity processed in the dataset.	Amount_NGN
Unique Digital Customers	Distinct customers with at least one transaction through a defined digital channel.	Shows breadth of digital engagement rather than only transaction volume.	Customer_ID, Channel
Channel Utilization	Transactions and/or unique customers by channel.	Shows which channels customers use and how usage is distributed.	Channel, Customer_ID
Device Utilization	Transactions and unique customers by device type.	Compares Android, iOS, Web Browser, POS Terminal and ATM Terminal usage.	Device_Type, Customer_ID
Transaction Success Rate	Successful transactions ÷ total transactions × 100.	Provides an operational view of transaction processing performance.	Transaction_Status
Failure/Reversal Rate	(Failed + Reversed transactions) ÷ total transactions × 100.	Highlights transaction outcomes that may warrant operational investigation.	Transaction_Status
Risk Review Rate	Risk-reviewed transactions ÷ total transactions × 100.	Monitors the proportion of transactions carrying the synthetic review label.	Risk_Review_Flag

9.1 Initial transaction KPI snapshot

Metric	Initial value
Total transactions	12,000
Total transaction value	NGN 560,477,354.85
Average transaction value	NGN 46,706.45
Median transaction value	NGN 10,305.94
Successful transaction rate	90.47%
Risk-review rate	19.60%
Customers represented in transactions	1,500 of 1,500

10. Priority Analysis Areas

10.1 Device usage: Android vs iOS vs browser vs terminals

One priority analysis is to determine whether the customers generating digital activity are primarily using Android, iOS, web browsers or terminal-based channels. This will be measured in two ways: transaction volume and unique customers. This prevents frequent users from being mistaken for a larger customer base.

Device type	Transaction count	Unique customers
Android	4,797	1,431
iOS	2,920	1,294
POS Terminal	1,571	979
Web Browser	1,447	915
ATM Terminal	1,169	826
Missing	96	Not calculated

The initial profile shows Android with the highest transaction count among the listed device types. However, the final interpretation will also consider unique customers, transaction value and transaction outcomes before making a business statement.

10.2 Geographic concentration

A second priority is understanding where transaction activity is concentrated. The dataset provides eight city-level locations. Because there is no explicit rural/town/village variable, the analysis will compare the available locations rather than infer settlement types.

Location	Transaction count
Lagos	1,431
Abuja	1,461
Port Harcourt	1,520
Kano	1,526
Ibadan	1,550
Enugu	1,484
Kaduna	1,481
Benin City	1,451
Missing	96

11. Dashboard Wireframe

20 Sep 2026 ▼	Nigeria ▼	CUSTOMER SEGMENT ▼	CHANNEL ▼
12,000 TOTAL TRANSACTIONS	₦XX.XM TOTAL VALUE	1,XXX DIGITAL CUSTOMERS	XX.X% SUCCESS RATE
CUSTOMER & DIGITAL ADOPTION Digital engagement by segment/age/income		CHANNEL USAGE Transactions by channel	

Preferred channel distribution Expected insight: differences in digital engagement.	Unique customers by channel Expected insight: volume vs customer adoption.
DEVICE USAGE Android • iOS • Web • POS • ATM Transaction volume vs unique customers Expected insight: dominant devices and reach.	GEOGRAPHIC ACTIVITY Transactions and value by supplied location Location distribution Expected insight: city-level geographic patterns.
TRANSACTION PERFORMANCE Success • Failed • Pending • Reversed Status by channel, device and transaction type	RISK REVIEW PATTERNS Risk-review rate and flagged transactions Patterns by channel, device, type and location

Dashboard section	KPI focus	Proposed visuals	Key filters
Page 1 — Executive Overview	KPI cards: Total Transactions, Total Value, Digital Customers, Success Rate, Failure/Reversal Rate, Risk Review Rate.	Trend by date; transaction mix; high-level channel distribution.	Date, location, customer segment, channel
Page 2 — Customer & Digital Adoption	Unique digital customers; average engagement score; customer counts.	Digital engagement by segment/age/income; preferred channel distribution.	Customer segment, age band, income band, account type, preferred channel
Page 3 — Channel & Device Usage	Transactions and unique customers by channel/device.	Channel and device comparisons; transaction volume vs unique customers.	Channel, device type, customer segment, location
Page 4 — Geographic & Transaction Analysis	Transactions and value by location and transaction type.	Location distribution; transaction type mix; value distribution.	Location, transaction type, date
Page 5 — Operational & Risk Patterns	Success, failure, reversal, pending and risk-review rates.	Status by channel/device/type; risk-review patterns.	Transaction status, risk flag, channel, device, transaction type

11.1 Major insights the dashboard should communicate

- How broadly customers are using digital banking, not just how many transactions are occurring.
- Which channels and devices account for the largest share of customer activity.
- Whether usage patterns differ by customer segment and demographic characteristics.
- Which locations have higher or lower transaction activity within the supplied dataset.
- Where transaction failures or reversals are more concentrated.
- Where the synthetic risk-review label is more common and therefore merits further investigation.

12. Technical Requirements & Tooling Approach

The Data Analytics assignment recommends Excel, SQL, Python/Pandas and Power BI, and specifically advises using tools appropriately rather than forcing every tool into the assignment. The proposed workflow is:

Tool	Planned use
Excel	Quick inspection, validation and manual review of suspicious or missing values.
Python	Main programming environment for the analytical workflow.
Pandas	Load, inspect, clean, merge, filter, group and aggregate the datasets.
NumPy	Optional numerical support where useful; not required for its own sake.
SQL	Demonstrate structured querying, filtering, grouping and KPI calculation where appropriate.
Jupyter Notebook	Document the analysis from data loading through findings in a reproducible sequence.
Power BI	Create the final interactive dashboard for management-facing communication.

12.1 Proposed analytical workflow

Stage	Activity
1. Inspect	Load the files and confirm records, fields, data types, categories and date ranges.
2. Validate	Check missing values, duplicates, identifiers, join integrity and suspicious values.
3. Prepare	Handle documented missingness and create analysis-ready variables.
4. Integrate	Join customer attributes to transactions using Customer_ID.
5. Explore	Use descriptive statistics, groupings and visualizations to identify patterns.
6. Measure	Calculate the approved KPIs and supporting metrics.
7. Communicate	Build the dashboard and write evidence-based insights.

13. Analytical Risks & Mitigations

Risk	Potential effect	Mitigation
Missing Device_Type/Location	Could distort device and geographic comparisons.	Quantify missingness, document it and apply a transparent treatment in Week 2.
Transaction volume mistaken for adoption	Frequent users may create many transactions.	Use both transaction counts and unique-customer counts.
Limited geographic granularity	The data does not identify villages, towns or rural/urban classification.	Restrict geographic conclusions to the eight locations supplied.
Synthetic data limitations	Patterns may not represent real-world banking behaviour.	Clearly label findings as analysis of synthetic educational data.
Risk label interpretation	Risk_Review_Flag is not a fraud determination.	Describe it as a synthetic educational label and use 'associated with' rather than causal language.
Skewed transaction amounts	A small number of high-value transactions can influence averages.	Report both mean and median and inspect distributions.

14. Conclusion

The Week 1 plan establishes a focused Data Analytics problem for FinTrust: understanding how customer and transaction behaviour varies across channels, devices, customer segments and locations, while also monitoring transaction value, operational outcomes and synthetic risk-review patterns.